

ECEN 704 VLSI Circuit Design Project

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Abstract—Briefly provide the description of the circuit designed and results obtained. This project focuses on the design of a telescopic operational amplifier developed using a mix of manual sizing and iterative simulation-based tuning. The amplifier was implemented in a 0.18 μm CMOS process, with all transistors designed using a consistent channel length of 270 nm as used in the schematic. The initial device dimensions were estimated using standard analog design equations and biasing conditions, and then refined through several rounds of SPICE simulations to improve gain, power efficiency, and stability. Throughout the process, performance metrics such as DC gain, phase margin, bandwidth, and common-mode behavior guided the adjustments in transistor widths and bias currents. The final amplifier meets the required specifications and shows better overall efficiency compared to a purely manual design approach. This combined methodology proved effective in reducing design time while achieving a well-balanced and reliable telescopic amplifier suitable for low-power analog applications.

I. INTRODUCTION

This project focuses on designing a high-performance fully-differential operational amplifier (op-amp) with an integrated common-mode feedback (CMFB) circuit, targeting a $1\text{ mm} \times 1\text{ mm}$ chip area and 1.8V supply voltage. The design combines traditional hand calculations for initial transistor sizing with iterative trial-and-error to ensure all transistors operate in saturation, achieving optimal performance

TABLE I. DESIGN SPECIFICATIONS

s.no	Design Spec Obtained	
	pret-layout	post-layout
DC GAIN	60.309	59.8
GBW	121MHz	121MHz
SLEW RATE	75.75V/us	75.75
NOISE	55u	55u
PHASE MARGIN	83.217 degrees	
POWER DISSIPATION		

Transistor	Length(n)	Width(u)	no of fingers
M1	270	84	84
M2	270	84	84
M3	270	42	42
M4	270	42	42
M5	270	84	84
M6	270	84	84
M7	270	84	84
M8	270	84	84
M9	270	12	12
M10	270	4	4
M11	270	4	4
M12	270	8	8
M13	270	8	8
CMFB CIRCUIT			
Transistor	length(nm)	Width	no of fingers
TP9	270	24	24
TP8	270	24	24
TN11	270	24	24
TN17	270	24	24
TN18	270	24	24
TN14	270	24	24
TN16	270	8	8
TN15	270	8	8

II. DESIGN PROCEDURE

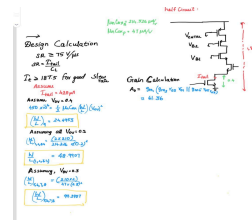
A. Circuit Topology

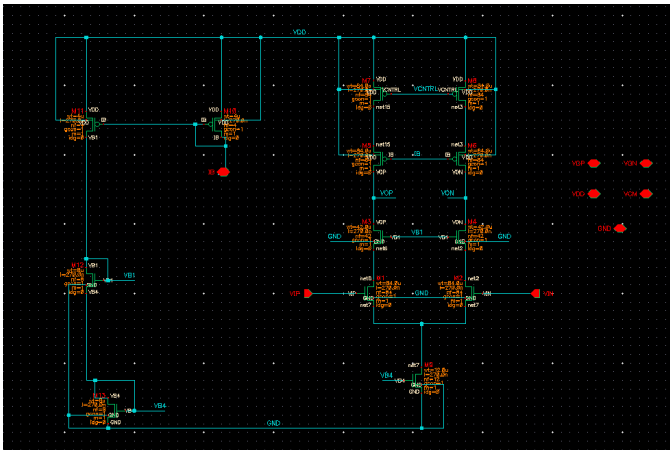
There are several fully-differential amplifier topologies available, including folded-cascode, two-stage, and telescopic architectures. For this design, the telescopic fully-differential amplifier was chosen to meet the required specifications efficiently. This architecture offers high open-loop gain and wide bandwidth while minimizing power consumption, making it suitable for high-speed, low-power applications.

One of the key reasons for choosing the telescopic design is its ability to avoid additional poles and zeros that can arise in multi-stage amplifiers, simplifying compensation and improving stability. Compared to folded-cascode and two-stage topologies, the telescopic amplifier provides superior gain-to-power ratio, lower input-referred noise, and better phase margin control, which are critical for high-performance analog designs. The main blocks planned for the design include: Differential input pair – for high gain and low input-referred noise. Current mirrors – for biasing and load implementation. Common-mode feedback (CMFB) circuit – to stabilize the output common-mode voltage. Output differential pair – to drive the load while maintaining linearity and speed.

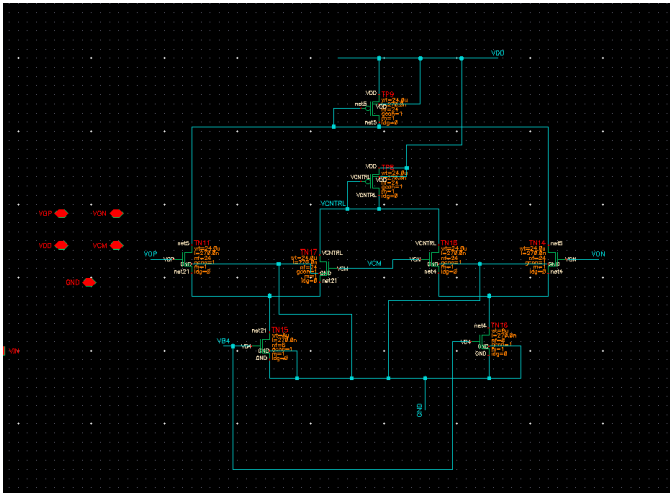
This architecture ensures that all specifications, including DC gain, GBW, slew rate, and noise requirements, are achievable within the given area and power constraints

B. Design Calculations

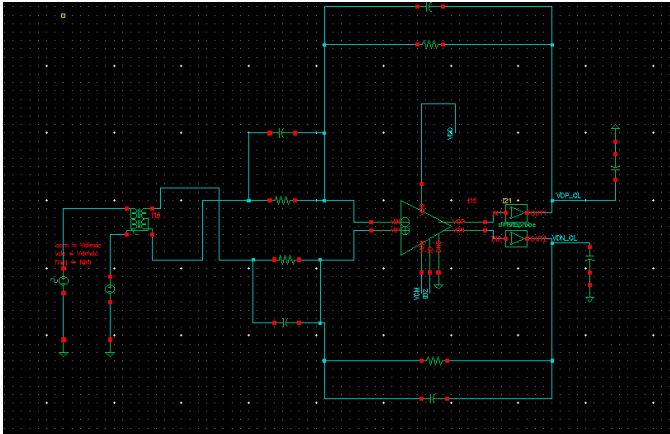




CMFB Schematic



Subcircuit Test bench Schematic:



Subcircuit for Stability Analysis:

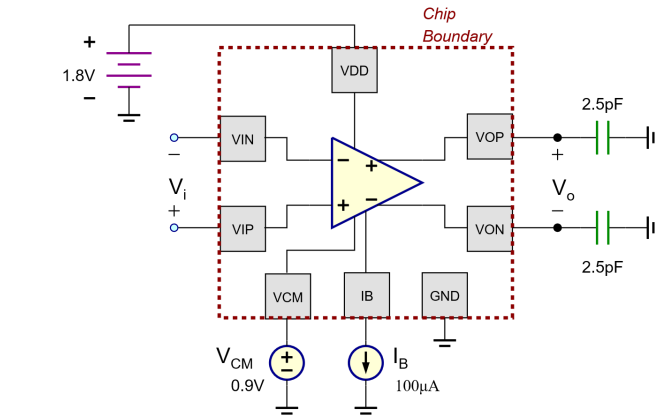
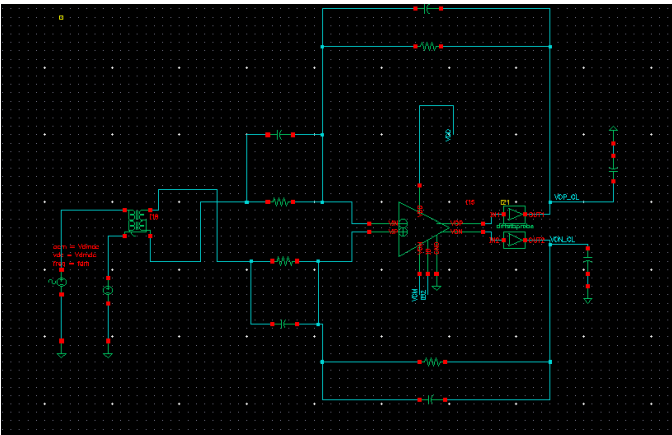
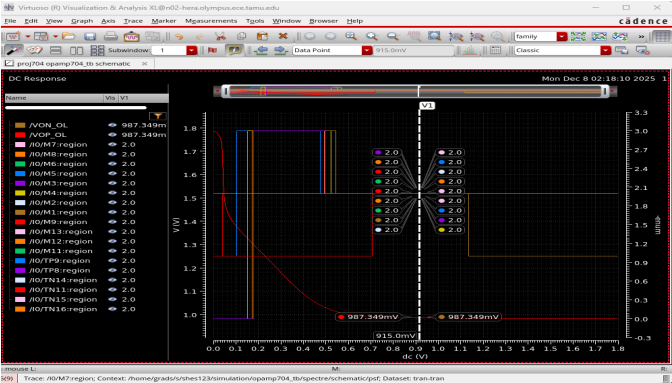


Fig. 1. Opamp testbench schematic.

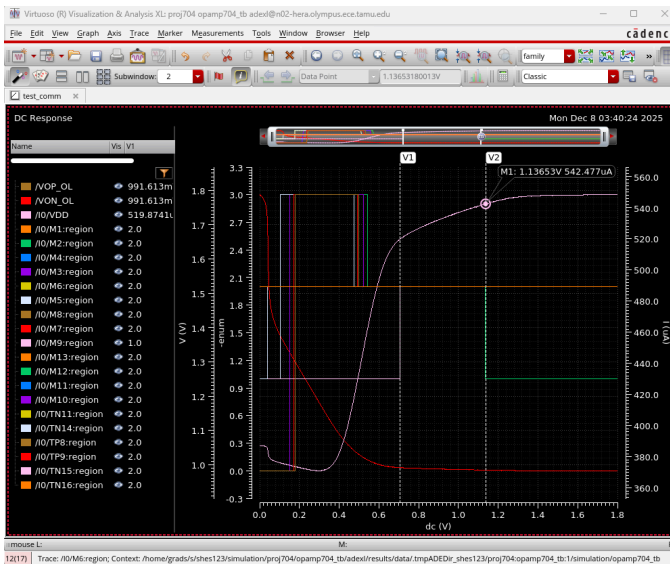
B. Simulation Results

Provide the testbench and simulation results showing that specifications are met. Label all plots to show measurements.

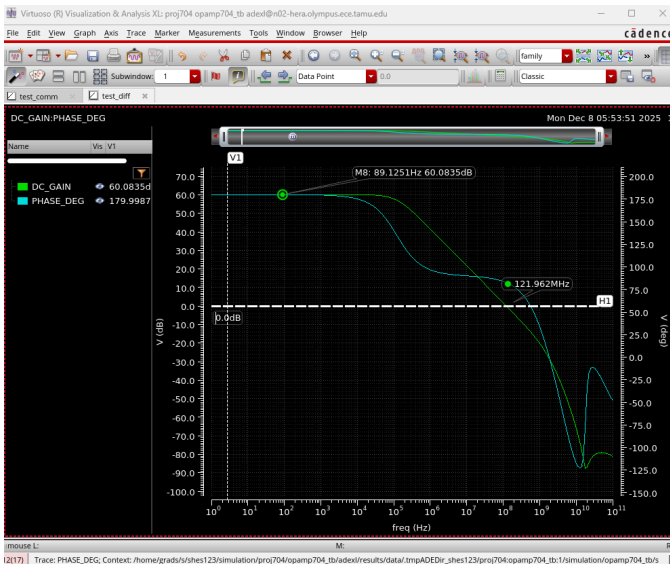
Graph showing that all transistors are in saturation:



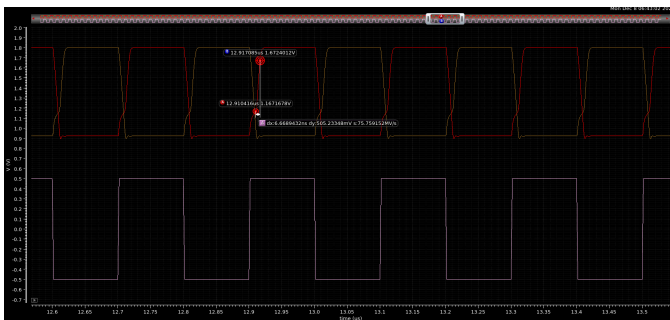
Graph showing the power consumption:



Dc gain and phase margin:



Output simulations for the slew rate:



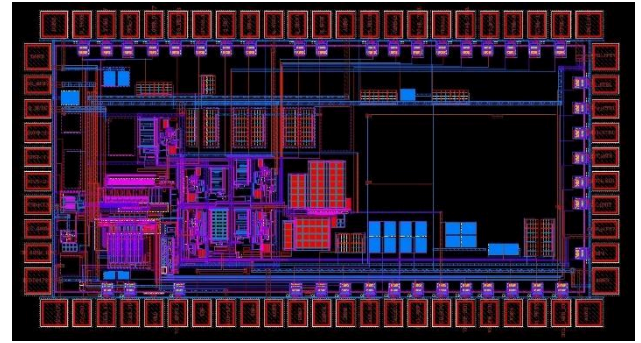
IV. CHIP LAYOUT

A. Layout Blocks

Display layout blocks, where input-output pins are labeled, and dimensions are shown.

B. Complete Chip

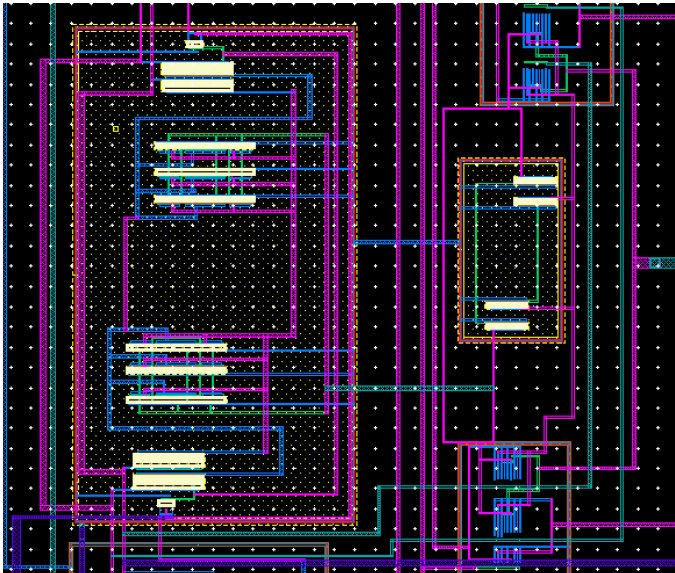
Show the complete chip layout including the pads, overall chip dimensions.



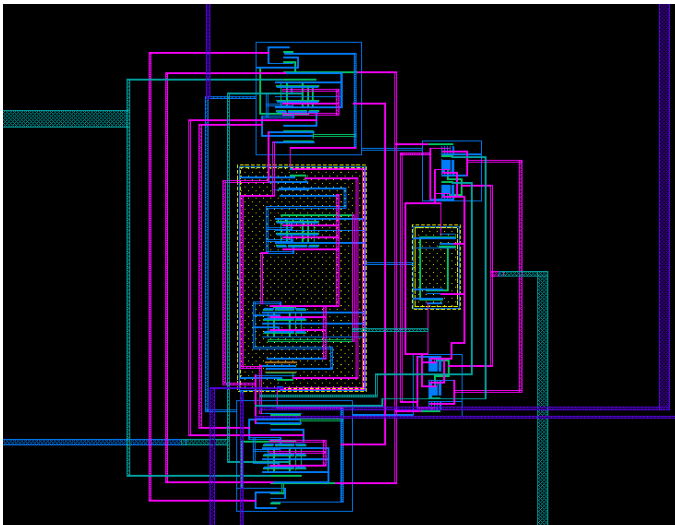
Chip Floor Plan:

change.log	NOTES.TXT
1	CMFB CIRCUIT :
2	TP16 (2) TP15 (4) TP16 (2)
3	TP14 (2) TP18 (4) TP14 (2)
4	TP14 (2) TP18 (4) TP14 (2)
5	TP14 (2) TP18 (4) TP14 (2)
6	TP11 (2) TP17 (4) TP11 (2)
7	TP11 (2) TP17 (4) TP11 (2)
8	TP11 (2) TP17 (4) TP11 (2)
9	TP8 (12)
10	TP9 (12)
11	TP9 (12)
12	TP8 (12)
13	TP11 (2) TP17 (4) TP11 (2)
14	TP11 (2) TP17 (4) TP11 (2)
15	TP11 (2) TP17 (4) TP11 (2)
16	TP14 (2) TP18 (4) TP14 (2)
17	TP14 (2) TP18 (4) TP14 (2)
18	TP14 (2) TP18 (4) TP14 (2)
19	TP11 (2) TP17 (4) TP11 (2)
20	TP16 (2) TP15 (4) TP16 (2)
21	
22	OPAMP CIRCUIT FLOOR PLAN :
23	M12 (4)
24	M13 (4)
25	M9 (4)
26	M1 (7) *M2 (7) M2 (7) *M1 (7)
27	M1 (7) *M2 (7) M2 (7) *M1 (7)
28	M1 (7) *M2 (7) M2 (7) *M1 (7)
29	M4 (21)
30	M3 (21)
31	M11M10 (2) M11
32	M6 (21)
33	M6 (21)
34	M5 (21)
35	M5 (21)
36	M7 (7) *M8 (7) M8 (7) *M7 (7)
37	M7 (7) *M8 (7) M8 (7) *M7 (7)
38	M7 (7) *M8 (7) M8 (7) *M7 (7)
39	
40	M7 (7) *M8 (7) M8 (7) *M7 (7)
41	M7 (7) *M8 (7) M8 (7) *M7 (7)
42	M7 (7) *M8 (7) M8 (7) *M7 (7)
43	M5 (21)
44	M5 (21)
45	M6 (21)
46	M6 (21)
47	M11M10 (2) M11
48	M3 (21)
49	M4 (21)
50	M1 (7) *M2 (7) M2 (7) *M1 (7)
51	M1 (7) *M2 (7) M2 (7) *M1 (7)
52	M1 (7) *M2 (7) M2 (7) *M1 (7)
53	M9 (4)
54	M13 (4)
55	M12 (4)
56	
57	
58	
59	

Closer look of PMOS:



CLOSER LOOK:



OVERALL LOOK OF THE LAYOUT WITH THYE I/O PADS:

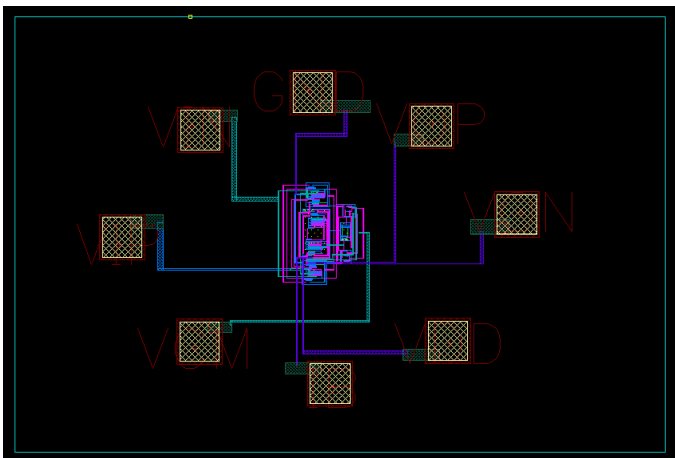
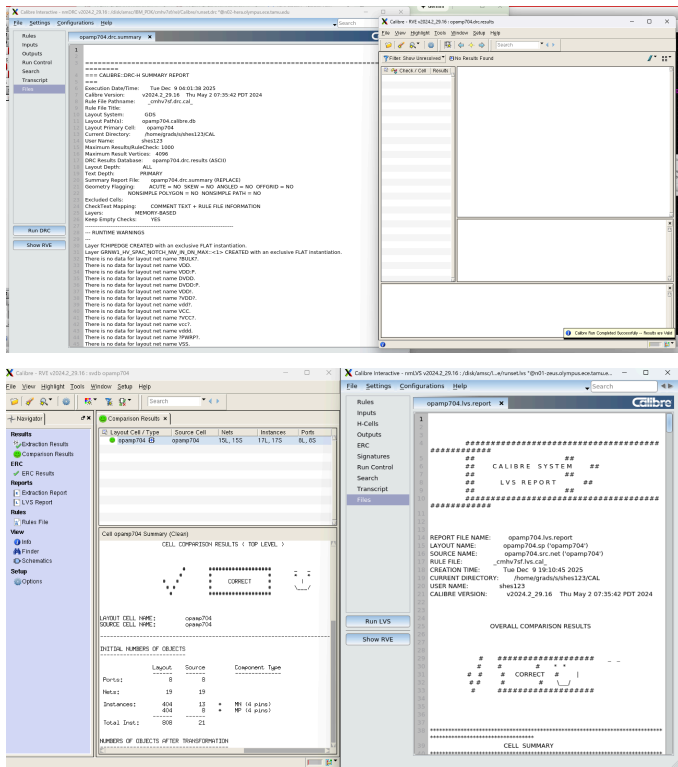


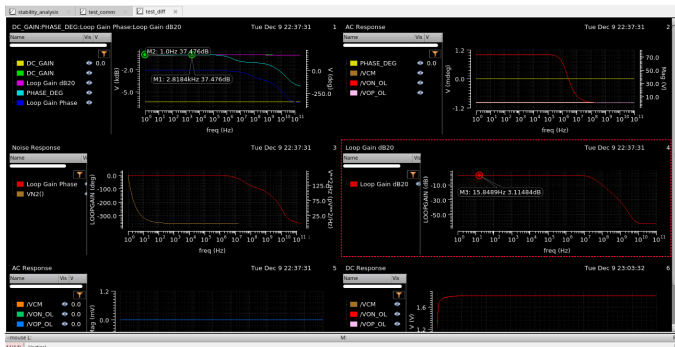
Fig. 2. Opamp Layout.

C. DRC and LVS Reports

Successfull DRC and LVS pasasing:



Provide post-layout simulation results. Label all plots to show measurements.



V. CONCLUSION

OP-AMP with the required specification is simulated and submitted

REFERENCES

[1] [1] A. Agarwal, *Design and analysis of CMOS telescopic operational amplifier*, M.S. Dissertation, Thapar Institute of Engineering Technology (Deemed University), Patiala, 2004.

[2] [2] K. Gulati and H.-S. Lee, "A high-swing CMOS telescopic operational amplifier," *IEEE Journal of Solid-State Circuits*, vol. 33, no. 12, pp. 2010–2019, 1998.

[3] [3] J. R. Angulo, M. Sawant, A. L. Martin, and R. G. Carvajal, "A power efficient and simple scheme for dynamically biasing cascode amplifiers and telescopic op-amps," *VLSI Journal*, vol. 41, pp. 239–243, 2008.

- [4] [4] P. E. Allen and D. R. Holberg, *CMOS Analog Circuit Design*, 2nd ed., Oxford, UK: Oxford University Press, 2002.
- [5] [5] L. C. Severo, "A framework for automatic scaling of analog integrated basic blocks," 2011, unpublished.
- [6] [6] J. R. Angulo, M. S. Sawant, R. Carvajal, and A. L. Martin, "New compact and power efficient dynamically biased cascode mirrors and telescopic op-amps," in *IEEE International Symposium on Circuits and Systems (ISCAS)*, Kos, Greece, May 21–24, 2006.